

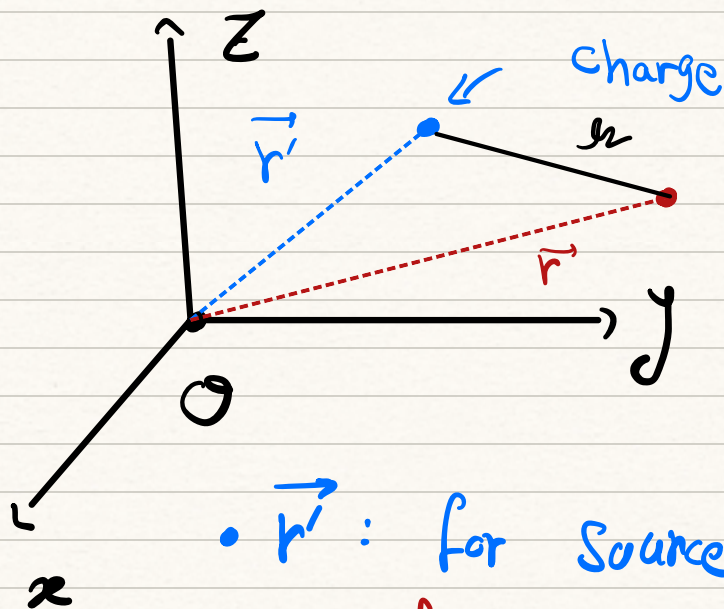
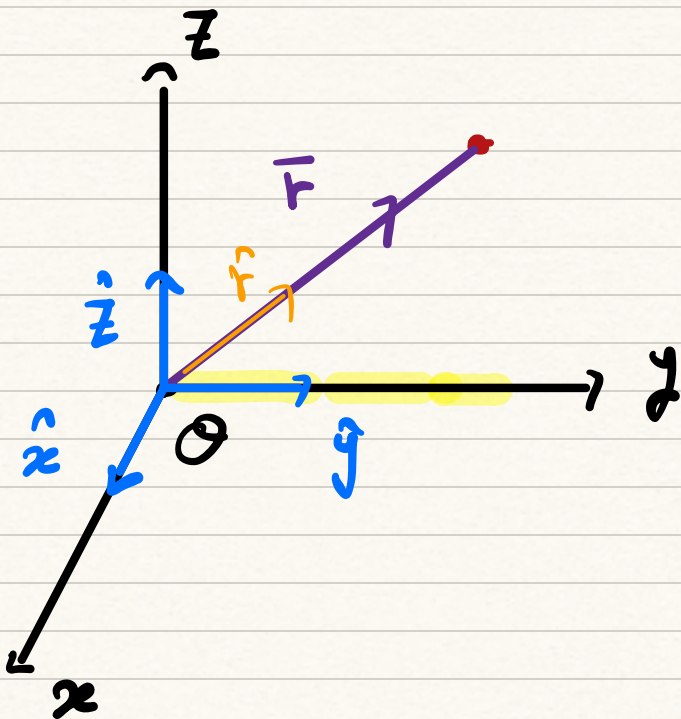
Chap 1: Vector Calculus

• Unit Vectors $|\hat{x}| = |\hat{y}| = |\hat{z}| = 1$

$$\vec{r} = x \hat{x} + y \hat{y} + z \hat{z}$$

$$r = |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$\hat{r} = \frac{\vec{r}}{r}$$



effect

• $\vec{r}'$: for Source Point

• $\vec{r}$: for Point of InterSet.

$\vec{r}$: Separation vector.

$$\vec{r} \equiv \vec{r} - \vec{r}'$$

Sec1 | Differential Operators

1) Gradient: ∇T

Apply on a scalar field to produce a vector
Points to the largest change in T

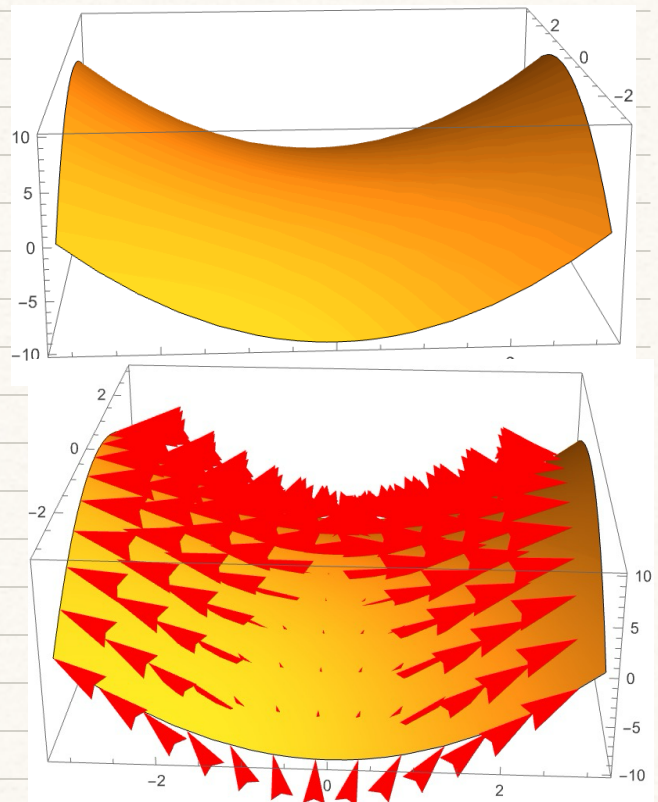
$$\nabla T = \left(\frac{\partial T}{\partial x}, \frac{\partial T}{\partial y}, \frac{\partial T}{\partial z} \right)$$

Exp 1.1) let $T = x^2 - y^2$, find ∇T

$$\frac{\partial T}{\partial x} = 2x ; \quad \frac{\partial T}{\partial y} = -2y$$

So

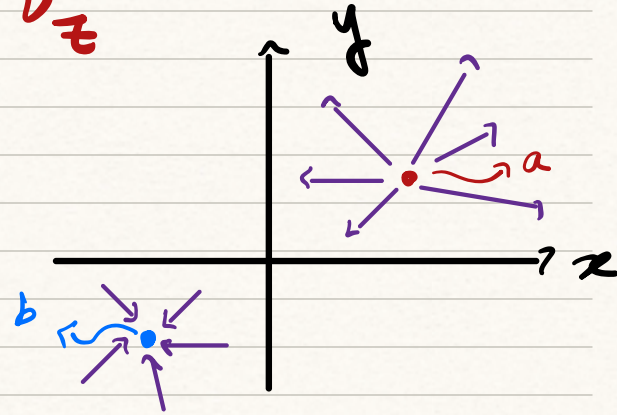
$$\nabla T = 2x \hat{x} - 2y \hat{y}$$



2) The Divergence: $\nabla \cdot \vec{v}$

Apply on a vector field to Produce a Scalar
measures the spreading of $\vec{v}$.

$$\nabla \cdot \vec{v} = \frac{\partial}{\partial x} v_x + \frac{\partial}{\partial y} v_y + \frac{\partial}{\partial z} v_z$$

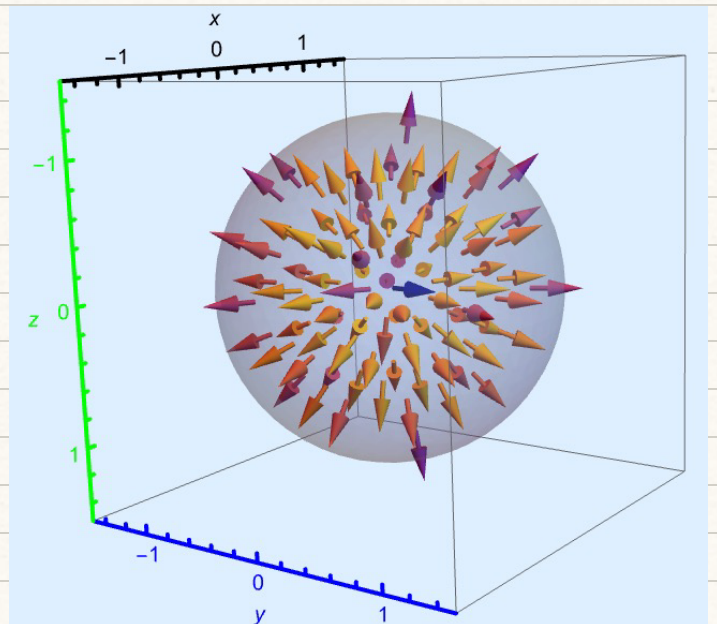


Exp 1.2) : $\vec{v} = e^{-(x^2+y^2+z^2)} (x \hat{x} + y \hat{y} + z \hat{z})$

Find $\nabla \cdot \vec{v}$

$$\begin{aligned} \frac{\partial v_x}{\partial x} &= \frac{\partial}{\partial x} \left(e^{-(x^2+y^2+z^2)} x \right) \\ &= -2x e^{-(x^2+y^2+z^2)} + e^{-(x^2+y^2+z^2)} \\ &= e^{-(x^2+y^2+z^2)} [1 - 2x] \end{aligned}$$

by Symmetry



$$\frac{\partial v_y}{\partial y} = e^{-(x^2+y^2+z^2)} [1-2y^2]$$

$$\frac{\partial v_z}{\partial z} = e^{-(x^2+y^2+z^2)} [1-2z^2]$$

$$\nabla \cdot \mathbf{v} = [3 - 2(x^2+y^2+z^2)] e^{-(x^2+y^2+z^2)}$$

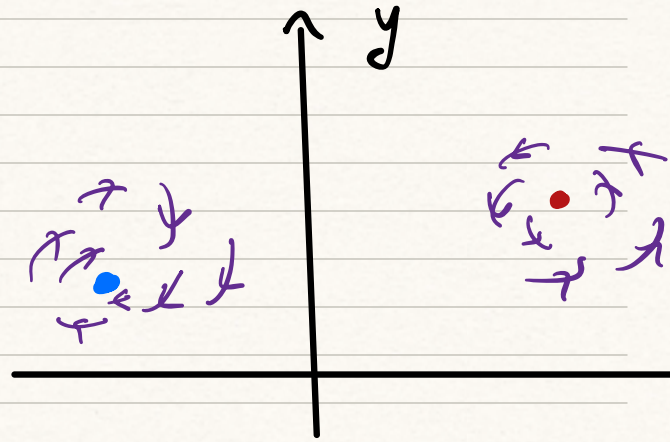
notice that $\nabla \cdot \mathbf{v} > 0$ for $x^2+y^2+z^2 < \frac{3}{2}$
inside a sphere of radius $\sqrt{\frac{3}{2}}$.

3) The Curl: $\nabla \times \vec{v}$

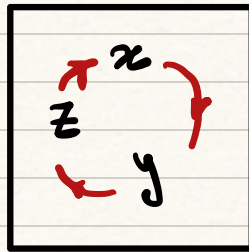
Apply on a vector field to Produce a Vector

Points to the axis of rotation of $\vec{v}$

$$\nabla \times \vec{v} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ v_x & v_y & v_z \end{vmatrix}$$



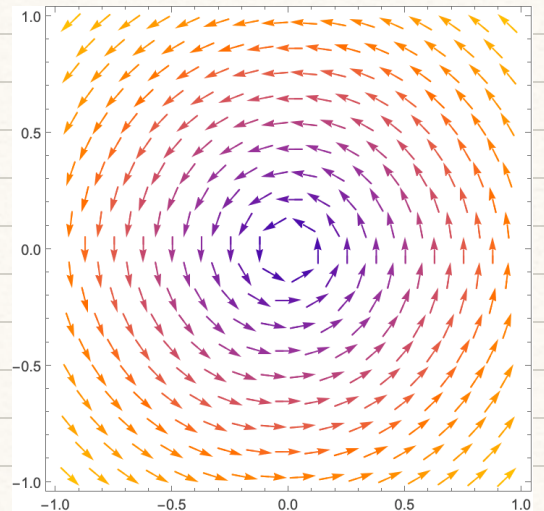
$$= (\partial_y v_z - \partial_z v_y) \hat{x} + (\partial_z v_x - \partial_x v_z) \hat{y} + (\partial_x v_y - \partial_y v_x) \hat{z}$$



EXP 1.3: $\vec{v} = -2y \hat{x} + 2x \hat{y}$

$$\nabla \times \vec{v} = [\partial_x (2x) - \partial_y (-2y)] \hat{z}$$

$$= (2 + 2) \hat{z} = 4 \hat{z}$$



Sec2

Fundamental Theorems

Bank account:

Initial $\text{£} 500$

$+\text{£} 50$

$-\text{£} 10$

$-\text{£} 20$

$+\text{£} 60$

$-\text{£} 30$

Final: $\text{£} 550$

Q: How much did we gain?

$$A1) +50 - 10 - 20 + 60 - 30 \\ = 50$$

$$A2) 550 - 500 = 50$$

$\Rightarrow$ The lesson is:

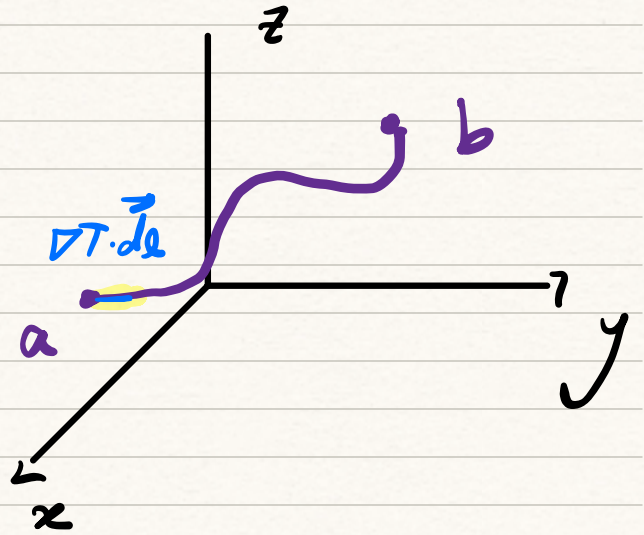
The sum of the small changes inside

$=$

The big change at boundary

1) Gradients:

$$\int \nabla T \cdot d\vec{l} = T(b) - T(a)$$



Exp 2.1) $T = x^2y + yz^2$, $a = (0,0,0)$, $b = (1,1,1)$

along $y = x^2$, $z = x$; $0 \leq x \leq 1$

Check $\int \nabla T \cdot d\vec{l} = T(b) - T(a)$

• Find LHS:

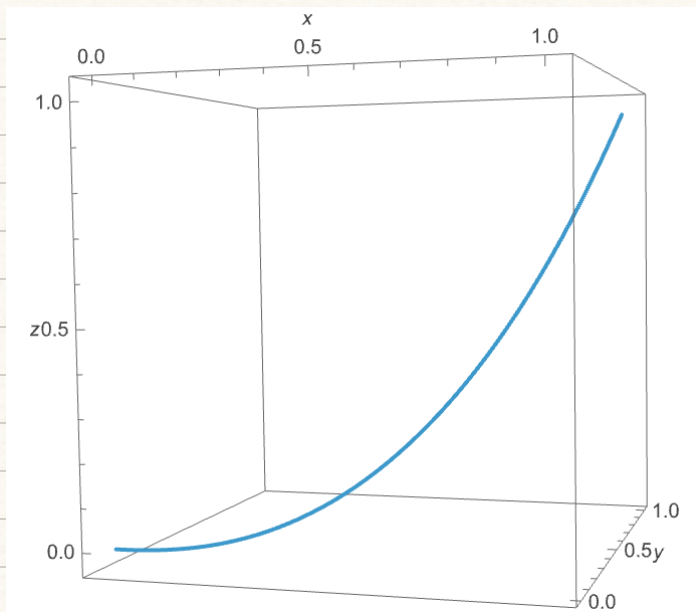
$$\nabla T = 2xy \hat{x} + (x^2 + z^2) \hat{y} + 2yz \hat{z}$$

$$d\vec{l} = dx \hat{x} + dy \hat{y} + dz \hat{z}$$

qs $z = x \Rightarrow dz = dx$ & $y = x^2 \Rightarrow dy = 2x dx$

$$\Rightarrow d\vec{l} = dx \hat{x} + 2x dx \hat{y} + dx \hat{z}$$

so $\nabla T \cdot d\vec{l} = 2xy dx + 2(x^2 + z^2) dx + 2yz dx$



$$= [2x(x^2) + 2(x^3 + x(x^2)) + 2(\underline{x^2})x] dx$$

$$= 8x^3 dx$$

$$\text{Thus } \int_{x=0}^{x=1} \nabla T \cdot d\mathbf{l} = \int_{x=0}^{x=1} 8x^3 dx = \frac{8}{4} x^4 \Big|_0^1 = 2$$

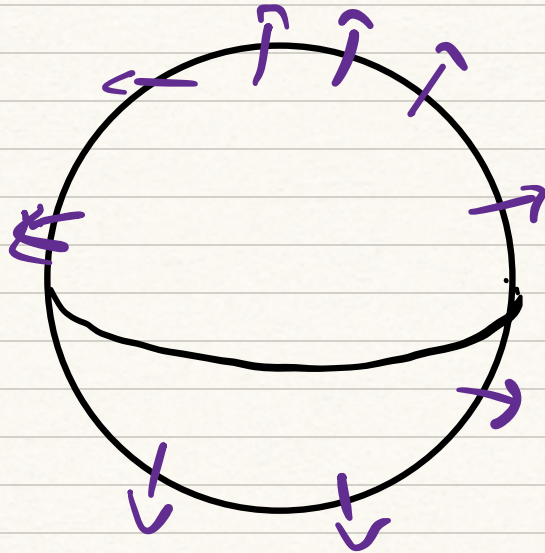
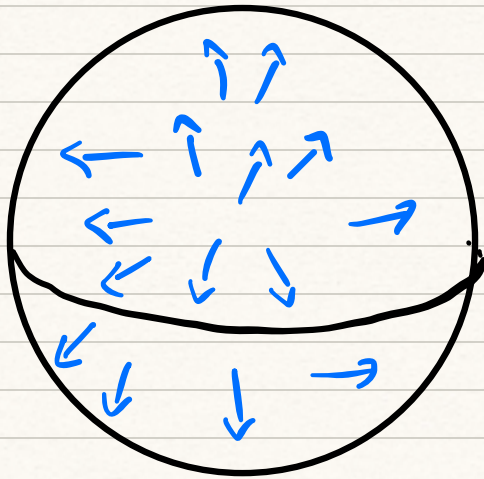
• Find RHS:

$$T(1,1,1) = 1^2 \cdot 1 + 11^2 = 2 ; T(0,0,0) = 0$$

$$\Rightarrow T(1,1,1) - T(0,0,0) = 2$$

$$\Rightarrow \text{LHS} = \text{RHS} \quad \checkmark$$

2) Divergences



$$\iiint_V \nabla \cdot \vec{v} \, d\tau = \oiint_S \vec{v} \cdot d\vec{a}$$

Sum of small divergences inside

=

One big divergences at the boundary, i.e.
Flux.

EXP2.2) $\vec{v} = x \hat{x} + 2y \hat{y} + z \hat{z}$

check: $\iiint \nabla \cdot \vec{v} = \oint \vec{v} \cdot d\vec{a}$

• Find LHS:

$$\nabla \cdot \vec{v} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(2y) + \frac{\partial}{\partial z}(z) = 1 + 2 + 1 = 4$$

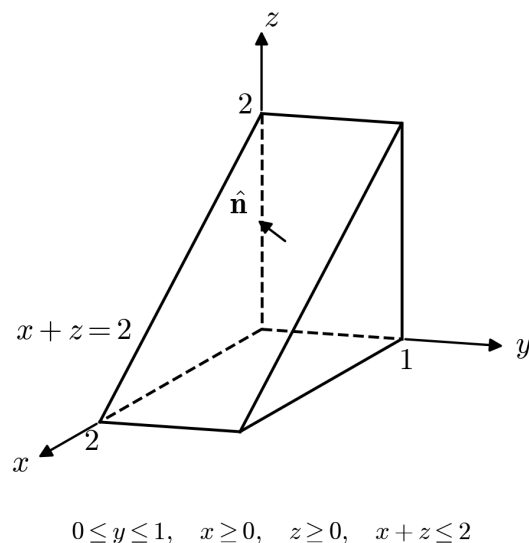
$$\iiint \nabla \cdot \vec{v} d\tau = 4 \int_0^1 \int_0^{2-x} \int_0^{2-x} dz dx dy = 4 \int_0^1 \int_0^2 (2-x) dx dy$$

$$= 4 \int_0^1 \left. \frac{(2-x)^2}{2} \right|_{x=2}^{x=0} dz = 4 \int_0^1 2 dz = 8$$

OR: $\iiint dz dx dy = \text{Volume of the Prism} =$
 $\frac{1}{2}(1)(2)(2) = 2$

• Find RHS:

Face I: $dx dz \hat{y}$ at $y = 1$



$$\mathbf{V} \cdot d\vec{a} = 2y \, dx \, dz = 2 \, dx \, dz$$

$$\Rightarrow \iint 2 \, dx \, dz = 2 \underbrace{\iint dx \, dz}_{\text{area of triangle}} = 2 \times \frac{1}{2} \times 2 \times 2 = \boxed{4}$$

Face II: $dx \, dz \, \hat{y}$ at $y=0$

$$\mathbf{V} \cdot d\vec{a} = 0 \Rightarrow \iint \dots = \boxed{0}$$

Face III: $dy \, dz \, (-\hat{x})$ at $x=0$

$$\mathbf{V} \cdot d\vec{a} = x \, dy \, dz = 0 \, dy \, dz \Rightarrow \iint \dots = \boxed{0}$$

Face IV: $dx \, dy \, (-\hat{z})$ at $z=0$

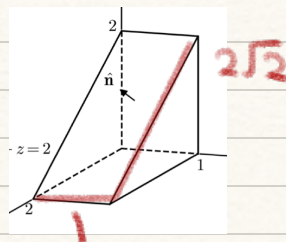
$$\mathbf{V} \cdot d\vec{a} = 0 \Rightarrow \iint \dots = \boxed{0}$$

Face V: slanted face $|x| + |z| = 2$

$$\vec{n} = 1\hat{x} + 1\hat{z} \Rightarrow \hat{n} = \frac{\hat{x} + \hat{z}}{\sqrt{2}}$$

$$\vec{v} \cdot \hat{n} = \frac{x+z}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

$$\text{so } \iint \vec{v} \cdot \hat{n} \, da = \sqrt{2} \iint da$$

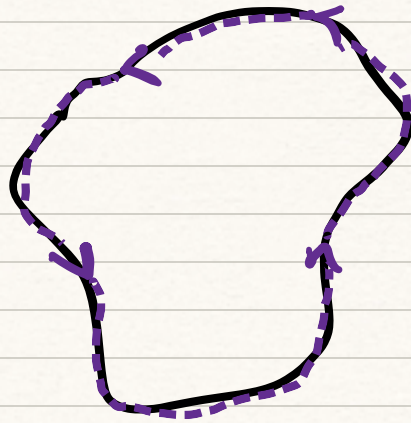
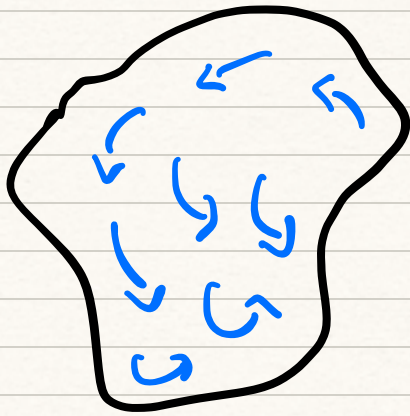


$$= 4$$

$$\text{Thus } \oint \vec{v} \cdot d\vec{a} = 4 + 0 + 0 + 0 + 4 = 8$$

$$\Rightarrow \text{LHS} = \text{RHS} \quad \checkmark$$

3) Stoke's Theorem



$$\iint_A (\nabla \times \vec{v}) \cdot d\vec{a} = \oint_P \vec{v} \cdot d\vec{l}$$

Sum of small circulations inside

=

one big circulation at the boundary

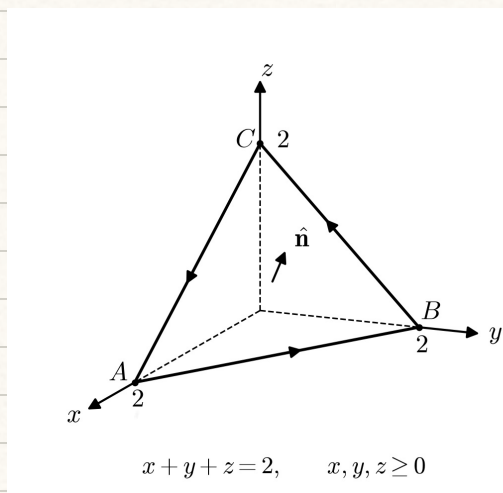
Gradients	$\int \nabla T \cdot d\vec{l}$	$T(b) - T(a)$	line integral $\Rightarrow$ diff end points
Div	$\iiint \nabla \cdot \vec{v}$	$\oiint \vec{v} \cdot d\vec{a}$	Volume integral = closed surface
Stoke's	$\iint (\nabla \times \vec{v}) \cdot d\vec{a}$	$\oint \vec{v} \cdot d\vec{l}$	Surface integral = closed path

Exp 2.3) $\vec{v} = z\hat{x} + x\hat{y} + y\hat{z}$

check $\iiint (\nabla \times \vec{v}) \cdot d\vec{a} = \oint \vec{v} \cdot d\vec{r}$

• Find LHS:

$$\nabla \times \vec{v} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ z & x & y \end{vmatrix} = \hat{x} + \hat{y} + \hat{z}$$



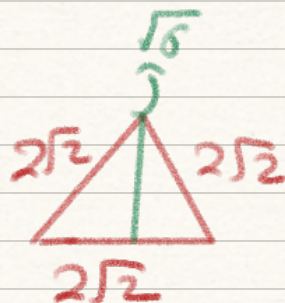
$$\vec{n} = 1\hat{x} + 1\hat{y} + 1\hat{z} \Rightarrow \hat{n} = \frac{1}{\sqrt{3}} (\hat{x} + \hat{y} + \hat{z})$$

$$(\nabla \times \vec{v}) \cdot d\vec{a} = (\nabla \times \vec{v}) \cdot \hat{n} da = \frac{1}{\sqrt{3}} (1+1+1) da$$

$$= \sqrt{3} da$$

$$\iiint (\nabla \times \vec{v}) \cdot d\vec{a} = \sqrt{3} \iiint da$$

area of



$$= \sqrt{3} \times \frac{1}{2} \times \sqrt{6} \times 2\sqrt{2} = \boxed{6}$$

Find RHS:

Path A $\rightarrow$ B: $y = 2 - x$; $z = 0$

$$d\vec{l} = dx \hat{x} - dy \hat{y}$$

$$\vec{v} \cdot d\vec{l} = z dx - x dy = -x dx$$

$$\int_2^0 -x dx = \int_0^2 x dx = \frac{1}{2} x^2 \Big|_0^2 = 2$$

Path B $\rightarrow$ C: $z = 2 - y$; $x = 0$

$$d\vec{l} = dy \hat{y} - dz \hat{z}$$

$$\vec{v} \cdot d\vec{l} = -y dy$$

$$\int_2^0 -y dy = \boxed{2}$$

Path C $\rightarrow$ A: $z = 2 - x$; $y = 0$. Same $\boxed{2}$.

$$\text{So } \int \vec{v} \cdot d\vec{l} = 2 + 2 + 2 = 6$$

$$\Rightarrow LSH = RSH \checkmark$$

Sec 3 | Coordinates Systems

1) Cartesian Coordinates:

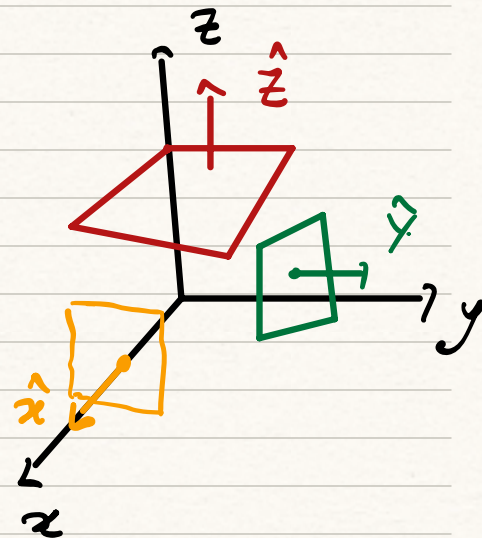
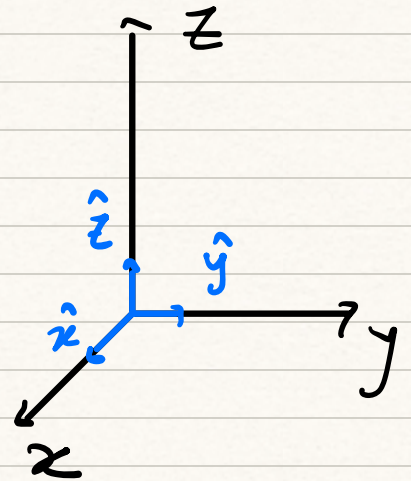
$$d\vec{l} = dx \hat{x} + dy \hat{y} + dz \hat{z}$$

$$d\vec{a} = dx dy \hat{z} \quad \text{Fix } z$$

$$d\vec{a} = dx dz \hat{y} \quad \text{Fix } y$$

$$d\vec{a} = dy dz \hat{x} \quad \text{Fix } x$$

$$dV = dx dy dz$$

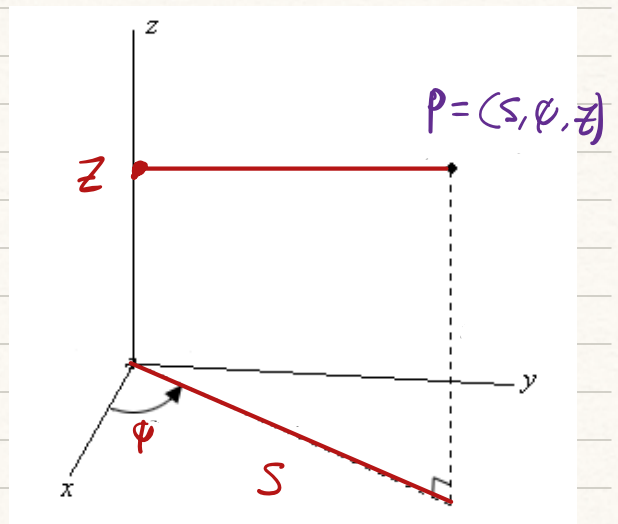


2) Cylindrical Coordinates:

$$x = s \cos \phi, \quad 0 \leq \phi < 2\pi$$

$$y = s \sin \phi$$

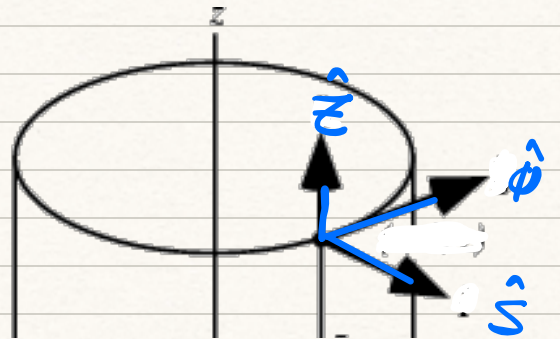
$$z = z$$



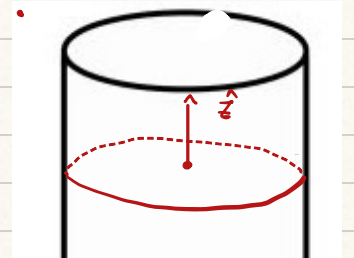
$$d\vec{r} = ds \hat{s} + s d\phi \hat{\phi} + dz \hat{z}$$

* why $s d\phi$?

$$s d\phi = s d\phi$$



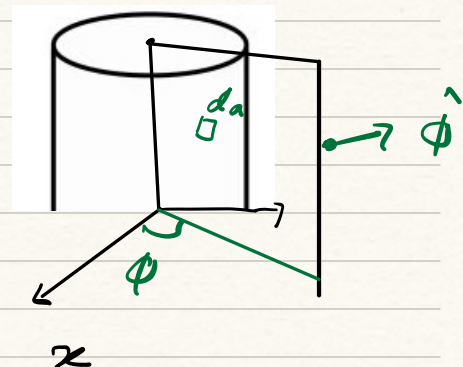
$$d\vec{a} = s d\phi ds \hat{z} \quad \text{Fix } z$$



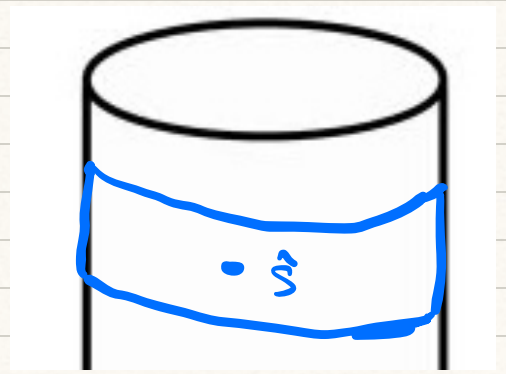
$$d\vec{a} = ds dz \hat{\phi} \quad \text{Fix } \phi$$

$$\hat{\phi} = -\sin \phi \hat{x} + \cos \phi \hat{y}$$

$\phi = 0$	$\phi = \frac{\pi}{2}$	$\phi = \pi$
$\hat{\phi} = \hat{y}$	$\hat{\phi} = -\hat{x}$	$\hat{\phi} = -\hat{y}$



$$d\vec{a} = s d\phi dz \vec{s} \quad \text{Fix } s$$



$$\Rightarrow d\tau = s ds d\phi dz$$

3) Spherical Coordinates:

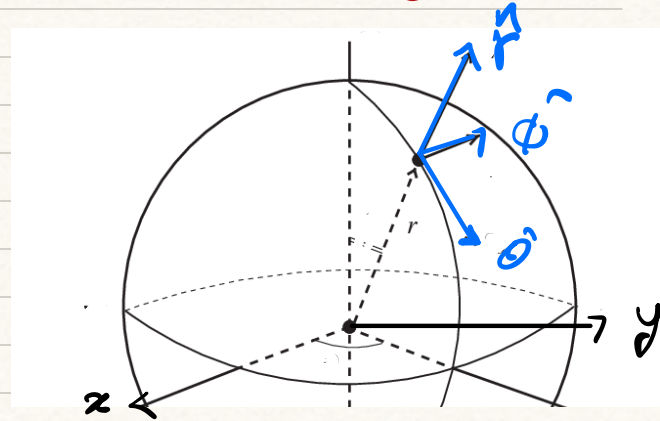
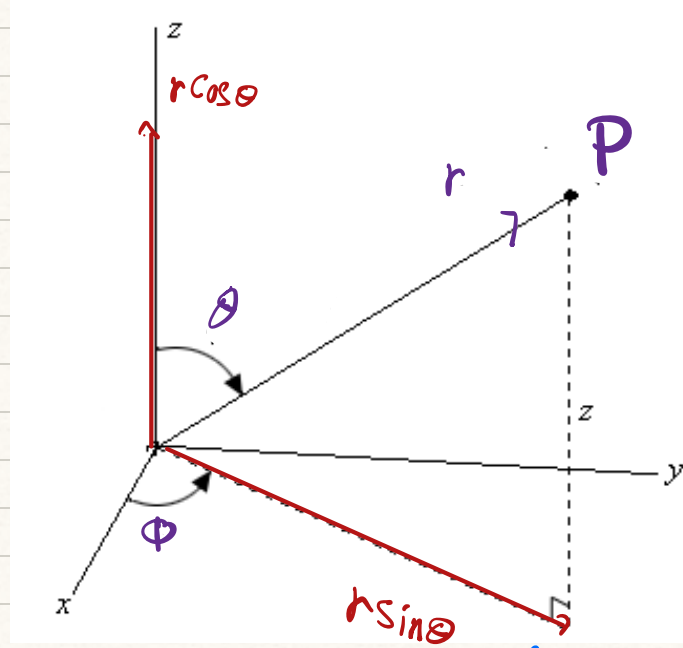
$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

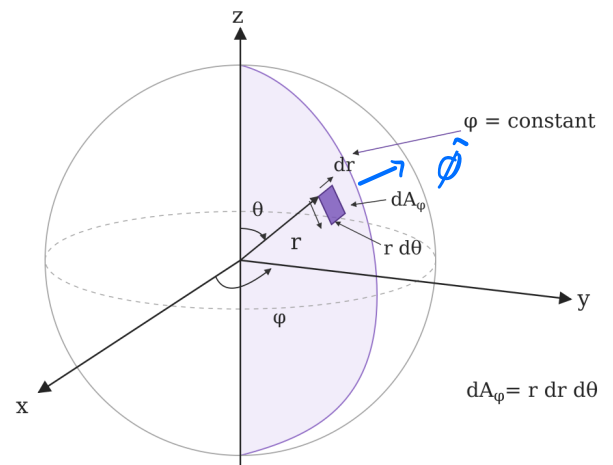
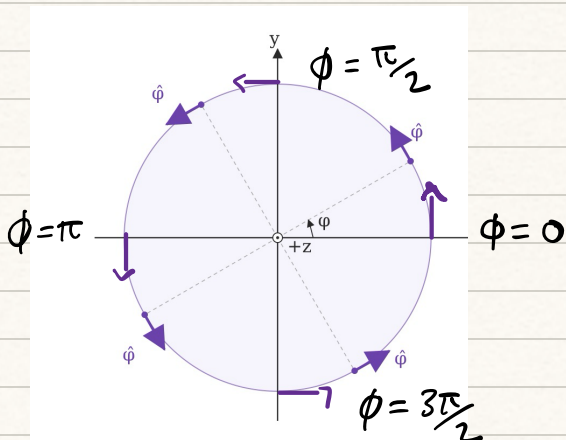
$$0 \leq \phi < 2\pi, \quad 0 \leq \theta \leq \pi$$

$$d\vec{r} = dr \hat{r} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi}$$

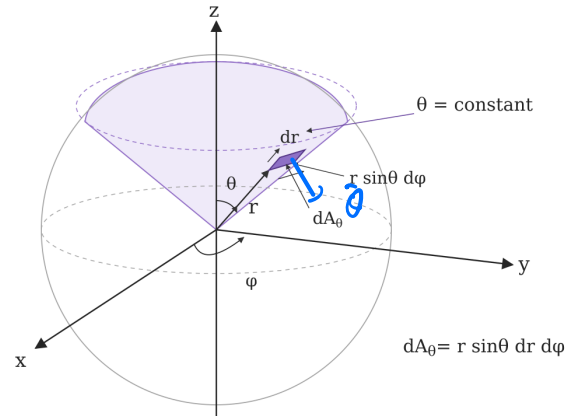


$$d\vec{a} = r dr d\theta \hat{\phi} \quad \text{Fix } \phi$$

$$\hat{\phi} = -\sin \phi \hat{x} + \cos \phi \hat{y}$$



$$d\vec{a} = r \sin \theta \, dr \, d\phi \quad \hat{\theta} \quad \text{Fix } \theta$$



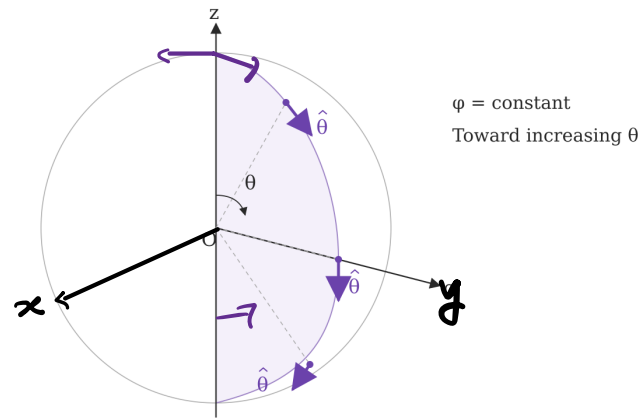
$$\hat{\theta} = \cos \theta \cos \phi \hat{x} + \cos \theta \sin \phi \hat{y} - \sin \theta \hat{z}$$

Case: $\theta = 0$

$$\phi = 0 \quad \hat{\theta} = \hat{z}$$

$$\phi = \frac{\pi}{2} \quad \hat{\theta} = \hat{y}$$

Case: $\theta = \frac{\pi}{2}$, $\hat{\theta} = -\hat{z}$



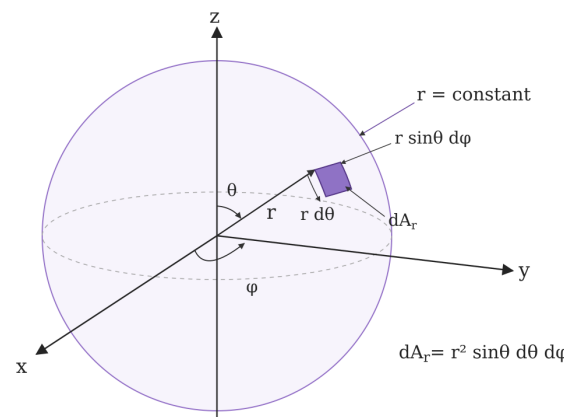
Case: $\theta = \pi$

$$\phi = 0 \quad \hat{\theta} = -\hat{z}$$

$$\phi = \frac{\pi}{2} \quad \hat{\theta} = -\hat{y}$$

$$d\vec{a} = r^2 \sin \theta \, d\theta \, d\phi \quad \hat{r} \quad \text{Fix } r$$

$$d\tau = r^2 \sin \theta \, dr \, d\theta \, d\phi$$



Sec 4 | Fundamental Theorems In

Curvilinear Coordinates:

• Cylindrical Coordinates:

$$\vec{\nabla} T = \left(\frac{\partial T}{\partial s} \right) \hat{s} + \frac{1}{s} \left(\frac{\partial T}{\partial \phi} \right) \hat{\phi} + \left(\frac{\partial T}{\partial z} \right) \hat{z}$$

$$\vec{\nabla} \cdot \vec{v} = \frac{1}{s} \frac{\partial}{\partial s} (s v_s) + \frac{1}{s} \frac{\partial}{\partial \phi} (v_\phi) + \frac{\partial v_z}{\partial z}$$

$$\vec{\nabla} \times \vec{v} = \left(\frac{1}{s} \frac{\partial v_z}{\partial \phi} - \frac{\partial v_\phi}{\partial z} \right) \hat{s} + \left(\frac{\partial v_s}{\partial z} - \frac{\partial v_z}{\partial s} \right) \hat{\phi} + \frac{1}{s} \left(\frac{\partial}{\partial s} (s v_\phi) - \frac{\partial v_s}{\partial \phi} \right) \hat{z}$$

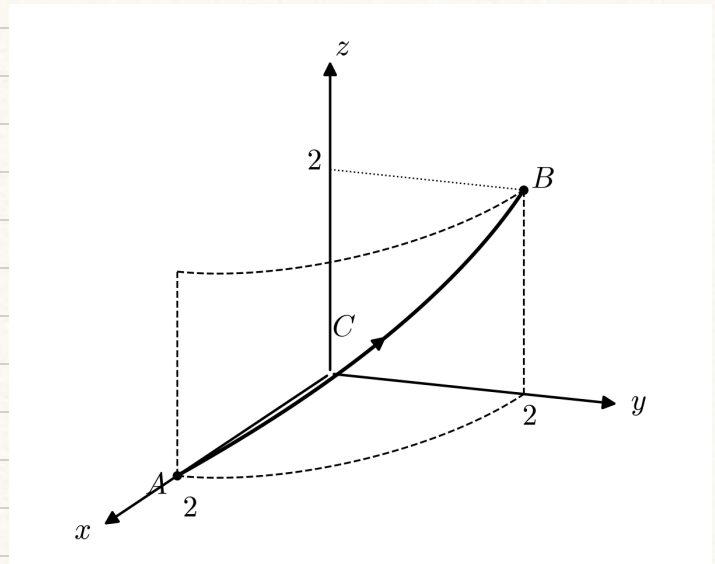
Ex 4.1) $T = s^2 \sin \phi + z^2$. $A = (2, 0, 0) \rightarrow B = (2, \pi/2, 2)$

Along the path

$$s = 2, \quad z = \frac{4}{\pi} \phi, \quad 0 \leq \phi \leq \frac{\pi}{2}$$

check $\int \nabla T \cdot d\vec{r} = T(B) - T(A)$

• Find LHS:



$$\frac{\partial T}{\partial s} = 2s \sin \phi ; \quad \frac{\partial T}{\partial \phi} = s^2 \cos \phi ; \quad \frac{\partial T}{\partial z} = 2z$$

$$\text{So } \nabla T = 2s \sin \phi \hat{s} + s \cos \phi \hat{\phi} + 2z \hat{z}$$

$$\vec{dl} = 2 d\phi \hat{\phi} + \frac{4}{\pi} d\phi \hat{z}$$

$$\nabla T \cdot \vec{dl} = \left[2s \cos \phi + \frac{8}{\pi} z \right] d\phi$$

$$= \left[2 \cdot 2 \cos \phi + \frac{8}{\pi} \left(\frac{8}{\pi} \phi \right) \right] d\phi$$

$$= \left[4 \cos \phi + \frac{32}{\pi^2} \phi \right]$$

$$\int \nabla T \cdot \vec{dl} = \int_0^{2\pi} \left(4 \cos \phi + \frac{32}{\pi^2} \phi \right) d\phi$$

$$= 4 + 4 = 8$$

• Find RHS: $T(2, \pi, 2) = 8, \quad T(0, 0, 0) = 0$

$$\Rightarrow \text{LHS} = \text{RHS} \quad \checkmark$$

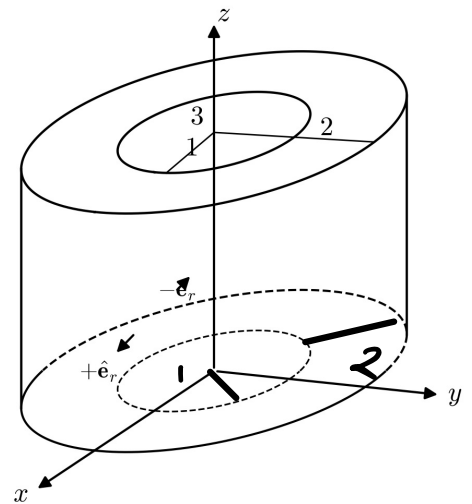
$$\text{Exp 4.2) } \vec{v} = s z \hat{s} + s^2 \hat{\phi} + z^2 \hat{z}$$

Check that

$$\iiint \nabla \cdot \vec{v} d\tau = \oiint \vec{v} \cdot d\vec{a}$$

• LHS:

$$\frac{\partial}{\partial s}(s^2 z) = 2s z; \quad \frac{\partial}{\partial \phi}(s^2) = 0; \quad \frac{\partial}{\partial z}(z^2) = 2z$$



$$\Rightarrow \nabla \cdot \vec{v} = 2z + 2z = 4z$$

$$\iiint \nabla \cdot \vec{v} d\tau = \int_0^3 \int_0^{2\pi} \int_1^2 4z s ds d\phi dz$$

$$= 4 \times 2\pi \int_0^3 z \left. \frac{s^2}{2} \right|_{s=1}^{s=2} dz = 4 \times 2\pi \times \frac{3}{2} \int_0^3 z dz$$

$$= 4 \times 2\pi \times \frac{3}{2} \times \frac{9}{2} = \boxed{54 \pi}$$

• RHS:

Face I: $s ds d\phi \hat{z}$ at $z=3$

$$\vec{V} \cdot d\vec{a} = z^2 s ds d\phi = 9 s ds d\phi$$

$$\Rightarrow 9 \int_0^{2\pi} \int_1^2 s ds d\phi = 9 \times 2\pi \times \frac{3}{2} = \boxed{27\pi}$$

Face II: $s ds d\phi (-\hat{z})$ at $z=0$

gives $\boxed{0}$

Face III: $s d\phi dz (\hat{s})$ at $s=2$

$$\vec{V} \cdot d\vec{a} = s z (s d\phi dz) = 4 z dz d\phi$$

$$\Rightarrow 4 \int_0^{2\pi} \int_0^3 z dz d\phi = 4 \times 2\pi \times \frac{9}{2} = \boxed{36\pi}$$

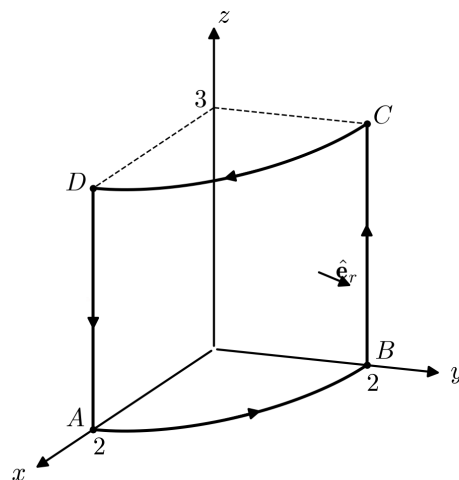
Face IV: $s d\phi dz (-\hat{s})$ at $s=1$

$$\vec{V} \cdot d\vec{a} = -z dz d\phi \Rightarrow \int (-) = \boxed{-9\pi}$$

So $\oint \vec{V} \cdot d\vec{a} = 54\pi \checkmark$

Ex 4.3) $\vec{v} = S(z+1) \hat{\phi} + S^2 \hat{z}$

check that $\iint_P \nabla \times \vec{v} \cdot d\vec{a} = \oint_P \vec{v} \cdot d\vec{r}$



LHS:

$$\frac{\partial v_\phi}{\partial z} = S ; \quad \frac{\partial v_z}{\partial S} = 2S$$

$$\frac{\partial}{\partial S} (S v_\phi) = 2S(z+1)$$

$$\Rightarrow \nabla \times \vec{v} = -S \hat{S} - 2S \hat{\phi} + 2(z+1) \hat{z}$$

$$d\vec{a} = S d\phi dz \hat{S} \text{ at } S = 2$$

$$= 2 d\phi dz \hat{S}$$

$$(\nabla \times \vec{v}) \cdot d\vec{a} = 2(-S) d\phi dz = -4 d\phi dz$$

$$\Rightarrow \int_0^3 \int_0^{2\pi} -4 d\phi dz = -8\pi$$

RHS:

Path $A \rightarrow B$: $s=2, z=0$

$$d\vec{l} = s d\phi \hat{\phi} = 2 d\phi \hat{\phi}$$

$$\vec{v} \cdot d\vec{l} = s(z+1) 2 d\phi = 4 d\phi$$

$$\Rightarrow \int_0^{\frac{\pi}{2}} 4 d\phi = \frac{\pi}{2} \cdot 4 = \boxed{2\pi}$$

Path $B \rightarrow C$, $s=2, \phi = \frac{\pi}{2}$

$$d\vec{l} = dz \hat{z}$$

$$\vec{v} \cdot d\vec{l} = s^2 dz = 4 dz$$

$$\int_0^3 4 dz = \boxed{12}$$

Path $C \rightarrow D$: $s=2, z=3$

$$d\vec{l} = 2 d\phi \hat{\phi}$$

$$\vec{v} \cdot d\vec{l} = 2 \cdot 2(3+1) d\phi = 16 d\phi$$

$$\int_{\frac{\pi}{2}}^0 r d\phi = \boxed{-8\pi}$$

Path D $\rightarrow$ A, $s=2, \phi=0$

$$d\vec{r} = dz \hat{z}$$

$$\vec{v} \cdot d\vec{r} = s^2 dz = 4 dz$$

$$\int_3^0 4 dz = \boxed{-12}$$

so

$$\oint \vec{v} \cdot d\vec{r} = 2\pi + 12 - 8\pi - 12 = \boxed{-6\pi}$$

$$\Rightarrow \text{LHS} = \text{RHS} \checkmark$$

• Spherical Coordinates:

$$\vec{\nabla} T = \frac{\partial T}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial T}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} \hat{\phi}$$

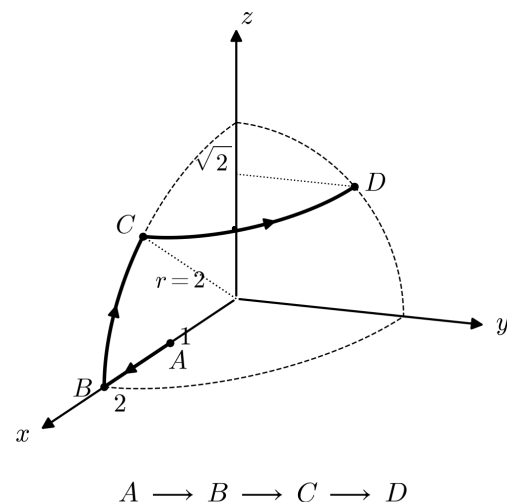
$$\vec{\nabla} \cdot \vec{v} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta v_\theta) + \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi}$$

$$\vec{\nabla} \times \vec{v} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta v_\phi) - \frac{\partial v_\theta}{\partial \phi} \right] \hat{r} + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{\partial}{\partial r} (r v_\phi) \right] \hat{\theta} + \frac{1}{r} \left[\frac{\partial}{\partial r} (r v_\theta) - \frac{\partial v_r}{\partial \theta} \right] \hat{\phi}$$

Exp 4.4) $T = r^2 \cos \theta + r \sin \theta \cos \phi$

check $\int \nabla T \cdot d\vec{l} = T(D) - T(A)$

$A = (1, \frac{\pi}{2}, 0), D = (2, \frac{\pi}{4}, \frac{\pi}{2})$



LHS: $\frac{\partial T}{\partial r} = 2r \cos \theta + \sin \theta \cos \phi ; \frac{\partial T}{\partial \theta} = -r^2 \sin \theta + r \cos \theta \cos \phi$

$\frac{\partial T}{\partial \phi} = -r \sin \theta \sin \phi$

So

$$\nabla T = (2r \cos \theta + \sin \theta \cos \phi) \hat{r} + (-r \sin \theta + \cos \theta \cos \phi) \hat{\theta} - \sin \phi \hat{\phi}$$

Path $A \rightarrow B$; $\phi = 0$, $\theta = \frac{\pi}{2}$

$$d\vec{l} = dr \hat{r}$$

$$\nabla T \cdot d\vec{l} = 1 dr \Rightarrow \int_1^2 1 dr = \boxed{1}$$

Path $B \rightarrow C$; $r = 2$, $\phi = 0$

$$d\vec{l} = 2 d\theta \hat{\theta}$$

$$\nabla T \cdot d\vec{l} = (-2 \sin \theta + \cos \theta) (2 d\theta)$$

$$2 \int_{\frac{\pi}{2}}^{\frac{\pi}{4}} (-2 \sin \theta + \cos \theta) d\theta = 2 \left[2 \cos \theta + \sin \theta \right]_{\frac{\pi}{2}}^{\frac{\pi}{4}}$$

$$= 2 \left[\frac{3\sqrt{2}}{2} - 1 \right] = \boxed{3\sqrt{2} - 2}$$

Path $C \rightarrow D$; $r=2$, $\theta = \frac{\pi}{4}$

$$d\vec{l} = r \sin\theta d\phi \hat{\phi} = 2 \frac{\sqrt{2}}{2} d\phi \hat{\phi} = \sqrt{2} d\phi \hat{\phi}$$

$$\nabla T \cdot d\vec{l} = -\sin\phi \sqrt{2} d\phi$$

$$\sqrt{2} \int_0^{\frac{\pi}{2}} -\sin\phi = \sqrt{2} \cos\phi \Big|_0^{\frac{\pi}{2}} = \boxed{-\sqrt{2}}$$

Thus

$$\int \nabla T \cdot d\vec{l} = 2\sqrt{2} - 1$$

$$T = r^2 \cos\theta + r \sin\theta \cos\phi$$

• RHS: $T(2, \frac{\pi}{4}, \frac{\pi}{2}) = 4 \cdot \frac{\sqrt{2}}{2} = 2\sqrt{2}$

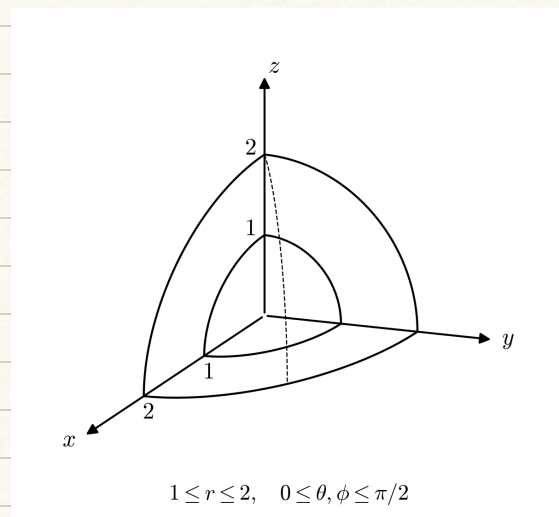
$$T(1, \frac{\pi}{2}, 0) = 1$$

$$\Rightarrow T(D) - T(A) = 2\sqrt{2} - 1$$

$$\Rightarrow LHS = RHS \quad \checkmark$$

Exp 4.5) $\vec{v} = r^2 \hat{r} + r \sin \theta \hat{\phi}$

check $\iiint \nabla \cdot \vec{v} \, d\tau = \oiint \vec{v} \cdot d\vec{a}$



LHS:

$$\frac{\partial}{\partial r}(r^4) = 4r^3; \quad \frac{\partial}{\partial \phi}(r \sin \theta) = 0$$

$$\nabla \cdot \vec{v} = 4r$$

$$\int_0^{\pi/2} \int_0^{\pi/2} \int_1^2 4r \, r^2 \sin \theta \, dr \, d\theta \, d\phi$$

$$= 4 \times \frac{\pi}{2} \int_0^{\pi/2} \int_1^2 r^3 \sin \theta \, dr \, d\theta$$

$$= 2\pi \cdot \frac{15}{4} = \boxed{\frac{15}{2} \pi}$$

RHS:

Face I: $d\vec{a} = r^2 \sin\theta d\theta d\phi \hat{r}$ at $r=2$

$$\vec{V} \cdot d\vec{a} = 4 \sin\theta d\theta d\phi$$

$$16 \int_0^{\pi/2} \int_0^{2\pi} \sin\theta d\theta d\phi = 8\pi$$

Face II: $d\vec{a} = r^2 \sin\theta d\theta d\phi (-\hat{r})$ at $r=1$

$$\vec{V} \cdot d\vec{a} = -\sin\theta d\theta d\phi$$

$$\Rightarrow \int \vec{V} \cdot d\vec{a} = \boxed{-\frac{\pi}{2}}$$

Face III: $d\vec{a} = r dr d\theta \hat{\phi}$ at $\phi = \frac{\pi}{2}$

$$\vec{V} \cdot d\vec{a} = r \sin\theta r dr d\theta = r^2 \sin\theta dr d\theta$$

$$\int_0^{\pi/2} \int_1^2 r^2 \sin\theta dr d\theta = \boxed{\frac{7}{3}}$$

Face IV: $d\vec{a} = r dr d\theta (-\hat{\phi})$ at $\phi = 0$

$$\vec{V} \cdot d\vec{a} = r^2 \sin\theta dr d\theta$$

$$\Rightarrow \int (\dots) = \boxed{-\frac{7}{3}}$$

Face V: $d\vec{a} = r \sin\theta dr d\phi \hat{\theta}$ at $\theta = \frac{\pi}{2}$

$$\Rightarrow \boxed{0}$$

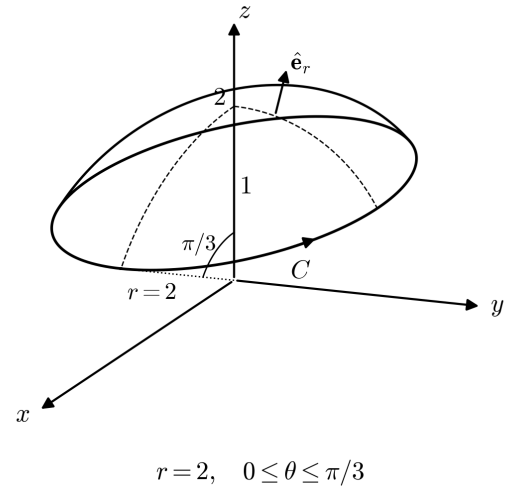
$$\Rightarrow \oiint \vec{V} \cdot d\vec{a} = 8\pi - \pi_2 = 15\pi_2$$

$$\Rightarrow \text{LHS} = \text{RHS} \checkmark$$

Exp 4.6)

$$\vec{v} = r \sin \theta \hat{\phi}$$

Show $\iint (\nabla \times \vec{v}) \cdot d\vec{a} = \oint \vec{v} \cdot d\vec{r}$



LHS:

$$\frac{\partial}{\partial \theta} (r \sin^2 \theta) = 2r \sin \theta \cos \theta$$

$$\frac{\partial}{\partial r} (r^2 \sin \theta) = 2r \sin \theta$$

$$\text{So } \nabla \times \vec{v} = 2 \cos \theta \hat{r} - 2 \sin \theta \hat{\theta}$$

$$d\vec{a} = r^2 \sin \theta d\theta d\phi \hat{r} = 4 \sin \theta d\theta d\phi \hat{r}$$

$$\nabla \times \vec{v} \cdot d\vec{a} = 8 \cos \theta \sin \theta d\theta d\phi$$

$$\iint (\nabla \times \vec{v}) \cdot d\vec{a} = 8 \int_0^{\pi/3} \int_0^{2\pi} \frac{1}{2} \sin 2\theta d\theta d\phi$$

$$= 8\pi \int_0^{\pi/3} \sin 2\theta d\theta = 8\pi \cdot \frac{1}{2} \cos 2\theta \Big|_0^{\pi/3}$$

$$= 8\pi \left[1 + \frac{1}{2} \right] = \boxed{6\pi}$$

• RHS:

$$d\vec{r} = dr \hat{r} + r d\theta \hat{\theta} + r \sin\theta d\phi \hat{\phi}$$

$$r=2, \theta=\frac{\pi}{3}$$

$$\Rightarrow d\vec{r} = 2 \cdot \sin \frac{\pi}{3} d\phi \hat{\phi} = \sqrt{3} d\phi \hat{\phi}$$

$$\vec{v} \cdot d\vec{r} = (r \sin\theta) \sqrt{3} d\phi = 2 \cdot \frac{\sqrt{3}}{2} \cdot \sqrt{3} d\phi = 3 d\phi$$

$$\Rightarrow \int \vec{v} \cdot d\vec{r} = \int_0^{2\pi} 3 d\phi = \boxed{6\pi}$$

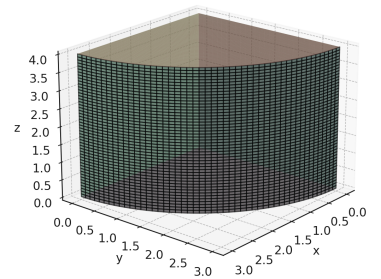
$$\Rightarrow \text{LHS} = \text{RHS} \quad \checkmark$$

Extra Problems:

P1) $\vec{v} = s^2 \cos \phi \hat{s} + s \sin \phi \hat{\phi} + z^2 \hat{z}$

Show $\iiint \nabla \cdot \vec{v} \, d\tau = \oiint \vec{v} \cdot d\vec{a}$

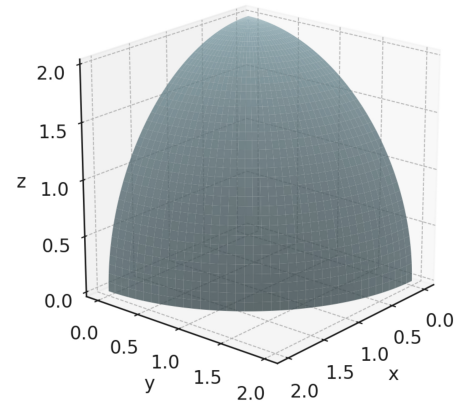
Quarter-cylinder region: $0 \leq s \leq 3$, $0 \leq \phi \leq \pi/2$, $0 \leq z \leq 4$



P2) $\vec{V} = r^2 \sin\theta \hat{r} + r \cos\theta \hat{\theta} + r \sin\theta \sin\phi \hat{\phi}$

Show $\iiint \nabla \cdot \vec{V} d\tau = \oiint \vec{V} \cdot d\vec{a}$

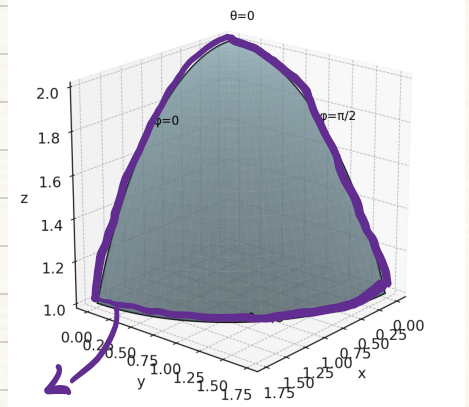
Spherical Octant ($0 \leq r \leq R$, $0 \leq \theta \leq \pi/2$, $0 \leq \phi \leq \pi/2$)



P3) $\vec{V} = \frac{1}{2} r \sin \theta \hat{\phi}$

Show $\oint_P (\nabla \times \vec{v}) \cdot d\vec{a} = \oint_P \vec{v} \cdot d\vec{x}$

Spherical rectangle: $R=2$, $0 \leq \theta \leq \pi/3$, $0 \leq \phi \leq \pi/2$



P

P4) $\vec{V} = x \hat{x} + y \hat{y} + z \hat{z}$

Find $\oiint \vec{V} \cdot d\vec{a}$

